# GOVERNMENT COLLEGE OF ENGINEERING, BARGUR

## DEPARTMENT OF ECE

## 25LPEV202 Professional Elective – II (HUMAN ASSIST DEVICES)

## INTERNAL ASSESSMENT TEST-I

**Year/Sem: III/V** **MAX. MARKS: 60**

**Date: 31.07.2026** **Duration: 90 Mins**

| S.No | PART –A(4x2=8 Marks) | Mark Obt. | COS | BTL |
|---|---|---|---|---|
| 1 | Why do we require heart lung machine? | 2 | CO1 | U |
| 2 | Write down the characteristics of blood pumps? | 2 | CO1 | R |
| 3 | Write down the different types of Artificial Kidneys. | 2 | CO3 | R |
| 4 | What is the use of Ventilator? | 2 | CO4 | U |
| | **PART-B (Answer all)** | | | **BTL** |
| 5 | Explain the working of H/L machine with its block diagram. | 18 | CO1 | U |
| 6 | Enumerate the different types of Oxygenators used in heart lung machine. | 16 | CO1 | U |
| 7 | Explain in detail the principle block diagram and working of haemodialyser. | 18 | CO3 | U |

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# Internal Assessment - I

1. A heart-Lung Machine is required to:
   * Maintain blood circulation during open-heart surgery.
   * Oxygenate the blood by adding oxygen and removing $CO_2$
   * Allow surgeons to stop the heart safely while operating.
   * Supply oxygen-rich blood to all body organs continuously.

2. Characteristics of blood pumps:
   * Should provide continuous and smooth blood flow.
   * Should cause minimum damage (hemolysis) to blood cells
   * Must avoid clot formation (thrombosis).
   * Should be reliable, safe and easy to sterilize.
   * Should have adjustable flow rate and pressure.
   * Should be biocompatible.

3. Types of Artificial kidneys:
   * Hemodialysis
   * Peritoneal Dialysis
   * Wearable Artificial kidney (WAK)
   * Implantable Artificial kidney (IAK)

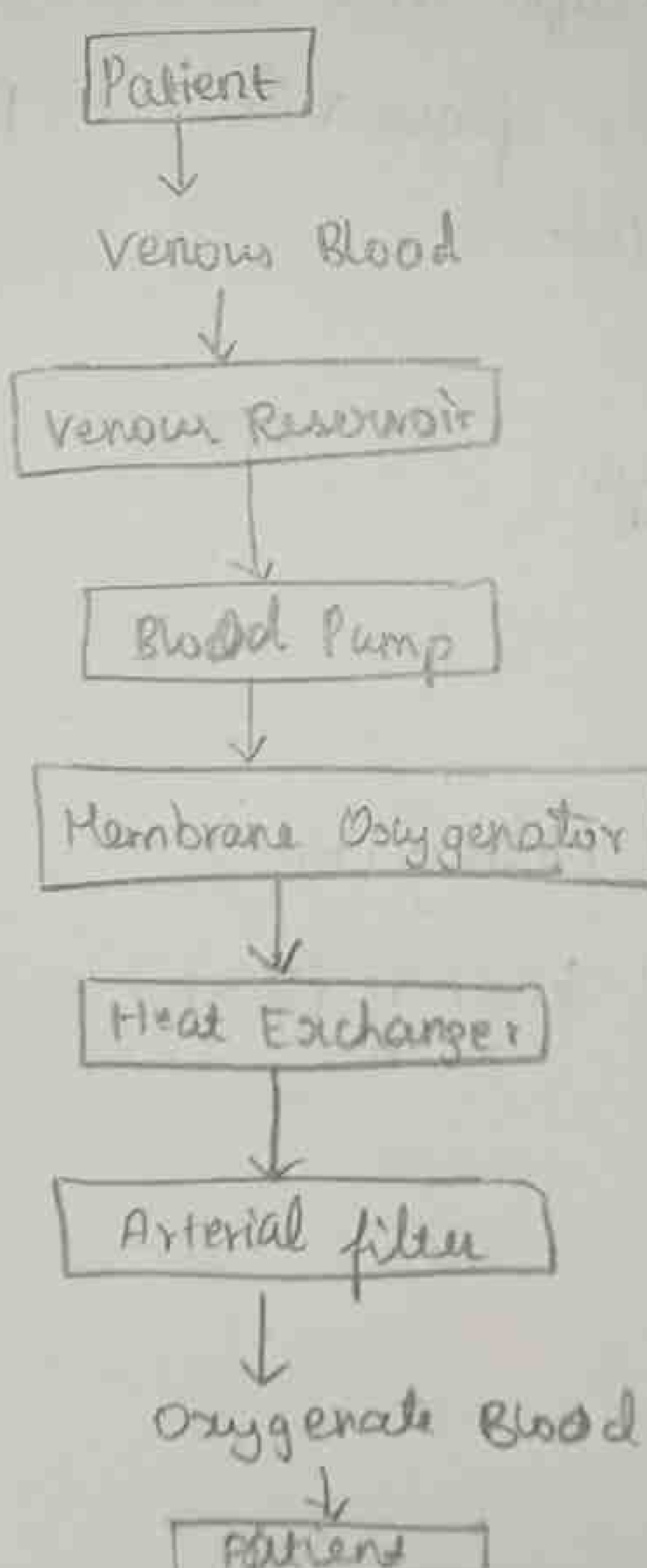
4. A Ventilator is a medical device that assists or completely controls breathing, Delivers oxygen-rich air to the lungs, removes carbon dioxide from the body and is used during surgery, respiratory failure and ICU treatment.

## Part-B

5. **Heart - Lung Machine:**

A- Heart - Lung Machine (Cardiopulmonary Bypass Machine) is a medical device that temporarily performs the functions of the heart and lungs during open-heart surgery. It pumps blood throughout the body and oxygenates it while the heart is stopped for surgery.

**Block diagram:**



Components:

1. Venous Reservoir:

* Collects deoxygenated blood from the patient
* Maintains sufficient blood volume
* Removes air bubbles before pumping.

2. Blood Pump:

* Pumps blood through the extracorporeal circuit
* Maintain proper blood flow and pressure.

* Types:
  - Roller pump
  - Centrifugal pump

3. Oxygenator:

* Acts as an artificial lung
* Removes carbon dioxide
* Adds oxygen to blood

4. Heat Exchanger:

* Maintain desired blood temperature
* Can cool or warm the blood during surgery.

5. Arterial Filter:

* Removes tiny air bubbles.
* Removes blood clots and debris
* Prevents embolism.

6. Tubing:

* Connects all components
* Made of biocompatible materials.

Working:

* Venous blood is drained from the patient's body into the venous reservoir.
* The blood pump draws blood from the reservoir.
* Blood enters the oxygenator where oxygen is added and carbon dioxide is removed.
* Blood passes through the heat exchanger where its temperature is adjusted.
* The arterial filter removes air bubbles and tiny particles.
* Oxygen-rich blood is pumped back into the patient's arteries.
* Thus, blood circulation and oxygen supply continue even when the heart is stopped.

Advantages:

* Enables open-heart surgery.
* Maintains blood circulation.
* Provides continuous oxygen supply.
* Controls body temperature

* Reduces risk during surgery.

Disadvantages:

* Expensive equipment
* Risk of blood clot formation
* Possibility of hemolysis
* Require trained personnel.

Applications:

* Coronary artery bypass surgery (CABG)
* Heart valve replacement
* Congenital heart surgery
* Heart transplantation.

⑤ Oxygenator:

An oxygenator is an artificial lung used in a heart-lung machine. It removes carbon dioxide from blood and supplies oxygen.

Types of oxygenators:

1. Bubble oxygenator:

Construction:

* Oxygen is bubbled directly into blood
* Contains bubble chamber and defoamer.

Working:

* Oxygen bubbles pass through blood.
* Oxygen dissolves in blood
* Carbon dioxide escapes through bubbles.
* Foam is removed before blood return.

Advantages:

* Simple construction
* Low cost
* Easy to operate

Disadvantages:

* Causes hemolysis
* Air embolism risk
* Less biocompatible.

2. Membrane Oxygenator:

Construction:

* Blood and oxygen are separated by a semipermeable membrane.

Working:

* Oxygen diffuses through membrane into blood
* Carbon dioxide diffuses from blood to gas side.
* Blood never directly contacts oxygen.

Advantages:

* Less blood damage
* Reduced embolism
* High efficiency
* Most commonly used.

Disadvantages:

* Costlier than bubble oxygenator.

3. Film Oxygenator:

Construction:

* Blood flows as a thin film over a large surface

Working:

* Thin blood film absorbs oxygen.
* Carbon dioxide escapes.

Advantages:

* Better gas exchange
* Less foaming

Disadvantages:

* Bulky
* Less commonly used.

4. Disc Oxygenator:

Construction:

* Rotating discs are partially immersed in blood.

Working:

* Thin blood film forms on rotating discs
* Oxygen enters blood
* Carbon dioxide leaves blood.

Advantages:

* Efficient oxygen transfer.

Disadvantages:

* Large size
* Complex construction.

5. Screen Oxygenator:

Construction:

* Blood flows over mesh screens.

Working:

* Thin blood layer is exposed to oxygen.

Advantages:

* Large surface area
* Good gas exchange.

Disadvantages:

* Difficult to clean
* Not widely used today.

Advantages of Membrane Oxygenator:

* Excellent oxygen transfer
* Less hemolysis
* Safe
* High bio compatibility
* Reduced air embolism.

Conclusion:

The membrane Oxygenator is the most widely used oxygenator because it provides efficient gas exchange with minimum blood damage.

7. Hemodialyser:

A Hemodialyser (Artificial kidney) is a medical device that removes waste products, excess salts and excess water from the blood when the kidneys fail.

Principle:

Hemodialysis works on three principles:

1. Diffusion:

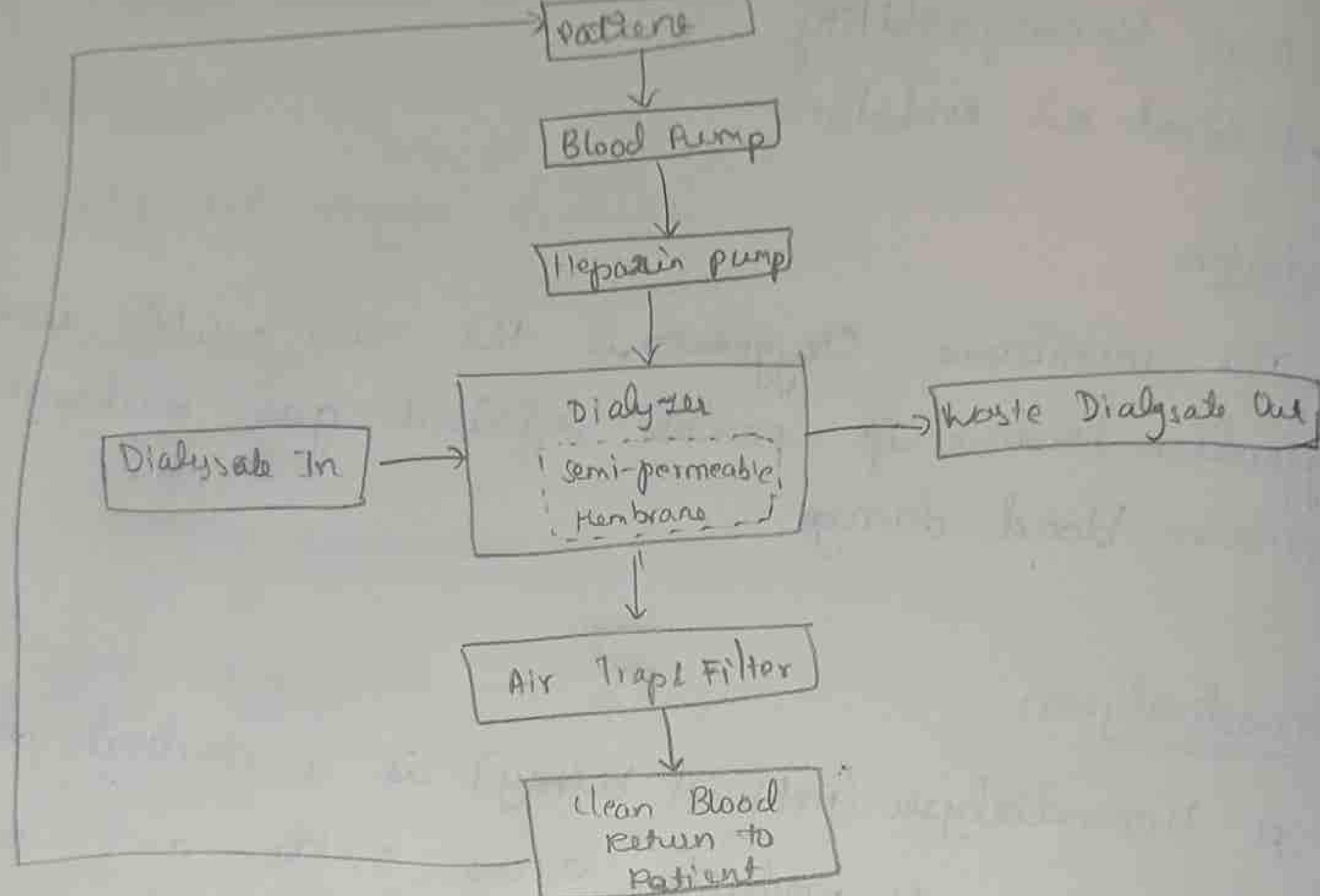
Waste substances like urea and creatinine move from blood (high concentration) to dialysate (low concentration).

2. Osmosis:

Water moves through a semipermeable membrane due to concentration difference.

3. Ultrafiltration:

Excess water is removed by applying pressure across the membrane




Components:

1. Blood pump:

Pumps blood through the dialyzer.

2. Heparin Pump:

Adds anticoagulant to prevent clotting.

3. Dialyzer:

Contains thousands of hollow fibers with semipermeable membranes.

4. Dialysate:

Special solution that removes waste while maintaing electrolyte balane.

5. Air trap:

Removes air bubbles from blood.

6. Pressure monitors:

Monitor blood pressure inside the circuit.

Working:

* Blood is withdrawn from the patient through vascular access.
* Heparin is added to prevent clot formation.
* Blood enters the dialyzer.
* Dialysate flows in the opposite direction.
* Urea, Creatine, excess salts and water pass through the membrane into the dialysate.
* Blood cells and protein remain in the blood because they are too large to cross the membrane.
* Air bubbles are removed by the air trap.
* Purified blood is returned safely to the patient's body.

Advantages:

* Removes waste products effectively.
* Maintains electrolyte balance.
* Removes excess water
* Improves quality of life
* Life-Saving treatment for kidney failure patients.

Disadvantages:

* Time-consuming (3-5 hours per session).
* Requires regular treatment (2-3 times per week).
* Risk of infection
* Blood pressure may decrease.
* Expensive long-term treatment.

Applications:

* Chronic kidney disease (CKD)
* Acute kidney failure
* Drug poisoning
* Severe electrolyte imbalance
* Fluid overload.